Predicting Attendance

Socioeconomic Hardship and Chronic Absenteeism in California School Districts (2024–25)

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Abstract

This study investigates the association of district-level poverty rates and chronic absenteeism within California public school districts during the 2024–2025 academic school year. We used administrative data from the California Department of Education ($n = 958$ districts with total enrollment minimum 50). We fitted a simple linear regression model relating Free or Reduced-Price Meal (FRPM) eligibility percentages (X) to district chronic absenteeism rates (Y). Ordinary Least Squares (OLS) estimation produced the fitted model $\hat{Y} = 7.7042 + 0.1855X$. The model shows a weak positive linear association ($R^2 = 0.1609$), with heteroskedasticity of the error variance being especially pronounced in higher socioeconomic hardship school districts. The unexplained variance in these school districts points to other factors other than FRPM that could be explored such as student health and wellness, housing stability, and transportation access. While district poverty is a statistically significant predictor when addressing chronic absenteeism ($p < 0.001$), policy interventions must go beyond socioeconomic support alone.

1 Introduction

To what extent does the percentage of socioeconomically disadvantaged students in a California school district predict its chronic absenteeism rate? As educators, we look for ways to help our students learn and succeed. We explore ways to improve pedagogy and content knowledge. We learn about different classroom routines or structures to implement, and often, we support our students in ways that go beyond the subject matter. However, none of this is possible or effective if the students themselves are not consistently in class. Thus, removing attendance barriers should be one of the primary aims of school districts, their respective school boards, and county leaders.

In a broader context, students regularly missing class is not necessarily a lack of motivation or a sign of defiance as one might might assume. Much research has been done on student attendance, and Attendance Works is one organization that

houses many of the seminal research and studies related to student attendance. When researching root causes influenced by family socioeconomic status, student absences could be closely tied to a family income, housing, transportation, child care, and health care. So as we continue to look for solutions to help our students get to class, exploring associations between poverty rates and chronic absenteeism rates can inform us if we need to go beyond basic socioeconomic interventions when addressing family and student needs.

For the analysis, we examined 2024–2025 Free or Reduced-Price Meal (FRPM) data (California Department of Education 2025c) and 2024–2025 chronic absenteeism data (California Department of Education 2025a) from the California Department of Education (CDE) to determine if student poverty rates could serve as a predictor of chronic absenteeism. Students who are chronically absent are those who are absent for 10 percent or more of the instructional days they were expected to attend. One key finding of this paper is that district-level FRPM rate could be used to predict absenteeism for California public school districts. However, in higher poverty districts, the variance is much greater. Therefore, when developing policies, interventions, and supports that address student attendance, higher poverty districts need to look for additional interventions in conjunction with socioeconomic supports. In this paper, Section 2 introduces the data used in this analysis, Section 3 describes the method, Section 4 shares the results, and Section 5 provides a discussion and possible next steps for further research.

2 Data

The analysis focuses on California public school districts ($n = 958$) in the database, in which we explored chronic absenteeism rate versus FRPM percent. For this purpose, we chose to focus on school districts with greater than 50 students enrolled for several reasons. First, school districts with less than 50 students do not typically represent standard, comprehensive K-12 school districts; these could be school districts composed of specialized alternative state schools who operate different guidelines when it comes to funding and attendance. Second, the CDE enforces privacy guidelines that protect student identities when the data comes from small populations; this data is often redacted. Third, small districts are subject to extreme statistical volatility. So when five students are chronically absent in a schools district of 10 students versus a school district of 1000 students, the smaller district becomes a point that could disproportionately influence the mode.

The CDE provides separate data for FRPM (California Department of Education 2025c) and chronic absenteeism (California Department of Education 2025a). FRPM data includes district code, school code, county name, district name, school name, K–12 enrollment, Free Meal Count, FRPM Count, and Percent Eligible FRPM. The student count is composed of students who are eligible for FRPM based on the same income eligibility as the National School Lunch

Program or who have foster, migrant, or homeless status. For FRPM, the district total for FRPM count and K–12 enrollment were not directly available in aggregate form, so data aggregation and manipulation were required. Absentee data includes district code, school code, county name, district name, school name, chronic absenteeism eligible cumulative enrollment, chronic absenteeism count, and chronic absenteeism rate. We calculated district totals by taking the sum of FRPM Count and the sum of K–12 enrollment across schools within each district to obtain new district totals, calculating the percentages from those aggregated sums. For chronic absenteeism, we aggregated the data to the district level and selected the overall aggregate reporting category. Lastly, we merged both data sets to obtain the finalized dataset that accounted for 958 matched cases. In the end, 14 school districts had missing or redacted absenteeism data due to state privacy redactions. Additionally, 42 school districts had no match in the FRPM dataset due to being a non-standard local educational agency (LEA) such as a county school program, juvenille hall program, or single room school.

Data is available at: https://github.com/kristophersonasuncion-code/MATH261A-project-1/tree/main

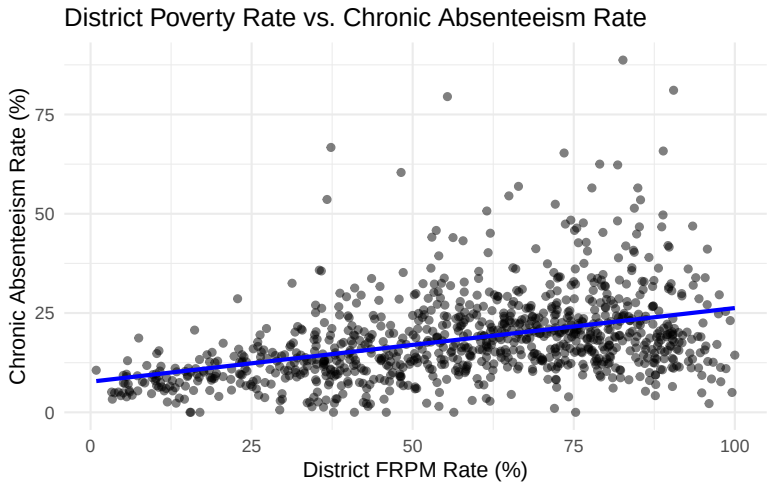


Figure 1: Scatter plot of District FRPM Rate (%) (x-axis) and Chronic Absenteeism Rate (%) (y-axis) with fitted linear regression model.

3 Methods

We adopt a simple linear regression model with chronic absenteeism rate as the response variable and poverty rate as the predictor variable. For a specific district i, the response variable Y_i represents the actual observed chronic absenteeism rate (percent) and the predictor variable X_i represents the FRPM rate (percent) for the ith California public school district in our dataset. The model can be

written as:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

where β_0 represents the predicted value of the chronic absenteeism rate (percent) when the FRPM rate is 0%. β_1 represents on average how much the chronic absenteeism rate (percent) increases for every 1% increase in FRPM rate.ε_i represents the error term and is assumed to follow an independent normal distribution with a mean of zero and constant variance σ^2 (Kutner et al. 2005, 40). We implemented this analysis using the R programming language (R Core Team 2024) and RStudio. The tidyverse meta-package (Wickham et al. 2019) was used for data manipulation. To import the data, we utilized readr and readxl (Wickham and Bryan 2023). The janitor package (Firke 2023) was used to standardize variable names between both datasets. The dplyr package (Wickham et al. 2023) allowed us to obtain district-level aggregate values, group individual schools into districts, calculate district FRPM percentages, remove duplications, and merge the datasets together by matching district codes. The linear regression model was fitted using base R's lm() function. The ggplot2 package (Wickham 2016) was used for data visualization to generate both the scatterplot with our fitted regression line and the residual plot.

4 Results

After running the simple linear regression, we obtain the following estimated response model:

$$\hat{Y} = 7.7042 + 0.1855X$$

The estimated slope $b_1 = 0.1855$ tells us that the chronic absenteeism rate (percent) increases by 0.1855 percentage points on average for every 1 percentage point increase in FRPM rate. The estimated intercept is $b_0 = 7.7042$ which means the predicted baseline value of chronic absenteeism is 7.7042% when the FRPM rate is 0%. In Figure 1, we see our model indicates a weak positive linear association between chronic absenteeism percent (Y) and FRPM rate (X). We used a two-sided t-test to compare the null hypothesis $H_0 : \beta_1 = 0$ against the two-sided alternative hypothesis $H_1 : \beta_1 \neq 0$. In this situation, the null hypothesis represents no linear association between district poverty rates and chronic absenteeism, while the alternative hypothesis asserts a statistically significant linear association. In order for this test to reject the null hypothesis at significance level α, we require $|t^*| > t_{1-\alpha/2,\, n-2}$, where t^* is our test statistic equal to $b_1/s\{b_1\}$ (Kutner et al. 2005, 47). With significance level $\alpha = 0.05$, estimated slope $b_1 = 0.1855$, standard error $s\{b_1\} = 0.0137$, and $df = 956$ degrees of freedom, the critical value is $t_{0.975,\, 956} = 1.960$. Since our observed t-statistic of 13.537 is significantly greater than 1.960, we reject the null hypothesis H_0. Additionally, we evaluate the p-value—the probability of observing data as extreme or more extreme than what was observed under H_0. If the p-value is below our chosen alpha, we reject the null hypothesis (Gao 2024). In this case, the p-value is less than 2.2×10^{-16}, which is far below 0.05. We reject H_0

and conclude that there is a statistically significant linear association between district poverty rates and district chronic absenteeism rates. Additionally, we found that for a 95% confidence interval for the slope estimate b_1 ranges from a lower bound of 0.159 to an upper bound of 0.212. This tells us the range that the true population slope could fall between which is fairly narrow since it is close to 0.1855. If the confidence interval were wider, it would signify that the true population slope could be much more smaller or larger.

Table 1: Simple Linear Regression Parameter Estimates and 95% Confidence Intervals

Parameter	Estimate	Lower 95% CI	Upper 95% CI
Intercept ($\hat{\beta}_0$)	7.704	6.010	9.398
FRPM Poverty Rate ($\hat{\beta}_1$)	0.186	0.159	0.212

To ensure these inferences are valid, we applied diagnostics on the residuals. First, Residuals vs. Fitted plot helps us assess linearity of the regression function as well as constancy of our error variance (Kutner et al. 2005, 106). While the plot shows a general linear association between district poverty and chronic absenteeism, we can see that the constancy of the error variance does not hold. This is shown by a megaphone shape starting to develop.

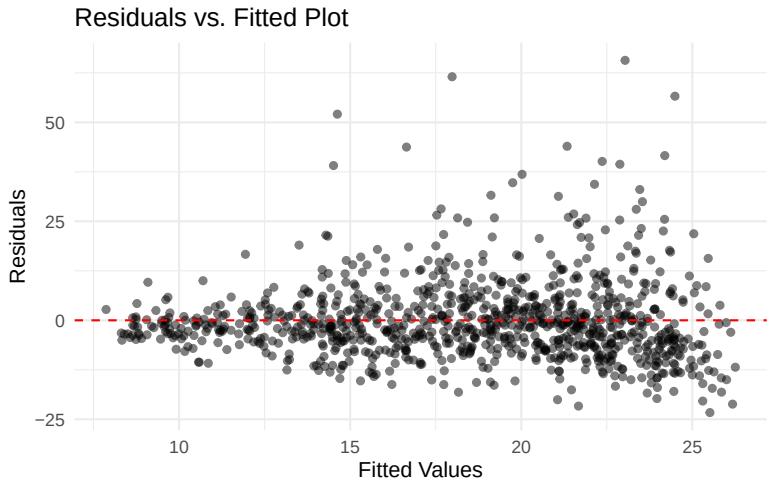


Figure 2: Scatter plot of fitted values (x-axis) and residuals (y-axis) for simple linear regression with Chronic Absenteeism Rate (%) as response and District FRPM Rate (%) as predictor.

Secondly, the Normal Q-Q plot helps us check for non-normality of errors (Kutner et al. 2005, 110). The plot below reveals the error term distribution is not

normal; it is skewed right. This indicates that higher poverty school districts have absenteeism rates that are more than we expected based on our model.

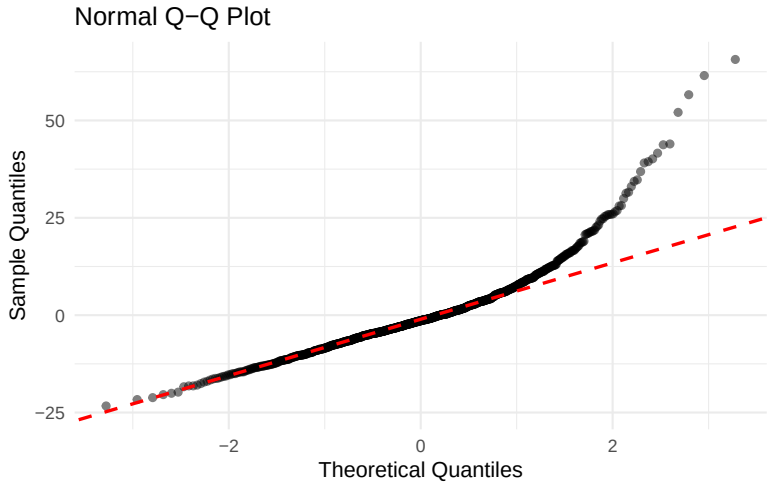


Figure 3: Normal Q-Q plot of model residuals to assess the normality assumption of the simple linear regression model.

While the residual Q-Q plot indicates right-skewness in the error distribution, sampling distribution of the slope estimate is approximately normal due to the large sample size ($n = 958$).

5 Discussion

When students are not consistently in class, they are missing opportunities for learning. If we want our students to succeed, we give them a better shot at success if they are in class. Removing barriers to learning starts with removing barriers to attendance. Therefore, exploring possible predictors of chronic absenteeism, such as district poverty rates, is worthwhile.

Despite the results that suggest a statistically significant linear association in this project, there are several weaknesses and limitations that we should address. One potential weakness of the fitted model is the heteroskedasticity of the error variance. Based on the Residuals vs. Fitted Plot (Figure 2), we observe an increasing vertical residual spread, demonstrated by a funnel shape, among districts with higher fitted values and higher FRPM rates. The heteroskedasticity creates problems that affect our error variance estimates (Kutner et al. 2005, 114). Because our standard error formula assumes homoskedasticity, our values for the standard error are not optimal estimates. Since our t-tests and p-values are dependent on that standard error calculation, we subsequently have less reliable t-tests, p-values, and confidence intervals.

Moreover, the the Residual vs. Fitted Plot indicates that prediction error is substantially higher in socioeconomically disadvantaged school districts than in wealthier school districts. Therefore, our model at higher FRPM rates may not be as reliable as it would be at lower FRPM rates. Another limitation is that the model yielded a coefficient of determination of $R^2 = 0.1609$, indicating that only 16.09% of the variation in chronic absenteeism rates is predictable from FRPM rates. Conversely, 83.91% of the variance in chronic absenteeism rates remains unexplained by this single predictor. This implies that other unobserved factors account for the vast majority of chronic absenteeism variation. Lastly, our sample could be subject to some sample bias. For our study, public school districts with total enrollment under 50 students were omitted due to data stability and confidentiality considerations. These omitted school districts may include Dashboard Alternative School Status (DASS) schools that serve high-risk youth dealing with housing instability or personal crises (California Department of Education 2025b). Because of the unique challenges these schools face, they may represent additional high-leverage outliers (high FRPM rate and high chronic absenteeism rate) that could influence the fitted model.

As for next steps, to address the heteroskedasticity of our model, we could explore the use of weighted least squares or transformations of the predictor variable (Kutner et al. 2005, 128). Additionally, future research could explore modeling chronic absenteeism specifically within districts where FRPM exceeds 50%. Because variance increases for districts with higher poverty rates, fitting a dedicated model on this range might yield deeper insights. We could also categorize schools by enrollment size intervals (e.g., 0–500, 500–1000, 1000–2000 students), as a district with 100 students differs fundamentally in context and administrative resources from one with 6,000 students. Additionally, categorizing school districts by locale (urban, suburban, or rural) would account for distinct economic and transportation barriers. Finally, for school districts with high FRPM rates, further research should explore student health and wellness access, affordable housing, public transportation infrastructure, and early childhood attendance patterns as critical factors driving chronic absenteeism.

For this study, we see that district poverty can help us make predictions about chronic absenteeism across California school districts. However, because district poverty accounts for only some the of attendance variance, educational leaders and policy makers cannot rely solely on FRPM rates to address chronic absenteeism. When determining interventions, especially in socioeconomically disadvantaged communities, we must look beyond economic relief alone to resolve systemic attendance deficiencies.

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